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# Safety-Aware Vehicle Routing for Quick-Commerce Last-Mile Delivery in Indian Cities: An $\varepsilon$-Constraint Multi-Objective Formulation with Calibrated Crash Risk

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**ABSTRACT** Quick-commerce platforms in India promise ten-minute delivery from neighbourhood "dark stores" to customers, putting persistent time pressure on a fleet of two-wheeler riders that must operate within an already-hazardous urban traffic environment. Conventional vehicle routing formulations minimise travel time and treat safety as a post-hoc concern. We propose a Safety-Aware Vehicle Routing Problem with Soft Time Windows (SA-VRPTW) that explicitly bounds per-route crash exposure and fleet-wide use of vulnerable residential roads via $\varepsilon$-constraints, while minimising a convex soft-time-window penalty as the primary objective. The crash risk per road segment is calibrated through a documented public-source chain (MoRTH 2022 + Mohan et al. 2017 + an OpenStreetMap structural proxy) that does not require access to the protected iRAD microdata. We implement three solvers under a unified evaluation contract: a piecewise-linear MILP, a genetic algorithm with Bellman–Ford optimal splitting, and an Adaptive Large Neighbourhood Search with Ropke–Pisinger adaptive operator weights. We benchmark on five Indian cities (Bengaluru, Delhi, Gurugram, Mumbai, Pune) with real Blinkit dark-store coordinates extracted from a public KML snapshot, sweeping instance sizes $N \in \{20, 50, 100, 200\}$ over ten seeds. The empirical study traces Pareto frontiers between $F_1$ delivery performance and per-route crash survival, quantifies the cost of the residential-road constraint, and verifies behavioural sensitivity through a Beta-prior compliance Monte Carlo. The contribution is a fully reproducible, calibration-honest framework that lets a quick-commerce operator dial the safety/efficiency trade-off explicitly rather than implicitly.

**INDEX TERMS** Adaptive large neighbourhood search, crash risk modelling, $\varepsilon$-constraint multi-objective optimisation, genetic algorithms, last-mile delivery, mixed-integer linear programming, quick-commerce, soft time windows, urban routing, vehicle routing problem.

## I. INTRODUCTION

QUICK-COMMERCE (q-commerce) platforms such as Blinkit, Zepto, and Instamart have rewired Indian urban delivery by promising arrival times measured in minutes rather than hours. The operating pattern is a network of small "dark stores" (also called depots or forward-deployed micro-warehouses), each serving a ∼2–3 km radius through a fleet of two-wheeler riders. Order batches are dispatched every few minutes, with each rider typically carrying one to two items per route to keep the delivery promise realistic. The combination of strict service-level agreements (SLAs), high dispatch frequency, narrow residential-street penetration, and a vulnerable rider population (helmeted but exposed two-wheelers in mixed traffic) creates a routing problem in which safety is not an optional refinement.

The classical Vehicle Routing Problem with Time Windows (VRPTW) formulation captures none of this. It minimises a weighted sum of travel time, distance, or vehicle count subject to capacity and time windows. Crash risk, when modelled, is typically embedded as an additional weighted term in the objective using a constant "risk-per-edge" coefficient that has no calibration story. Several works in safety-

aware routing penalise hazardous arcs implicitly via inflated travel times, which conflates two distinct phenomena (slow vs. dangerous) and gives the operator no explicit knob.

The contribution of this paper is a Safety-Aware VRPTW (SA-VRPTW) formulation that:

1) Treats crash exposure as a hard $\varepsilon$-constraint per route (survival probability $\geq e^{-\bar{R}}$) rather than as an objective component, so the safety/efficiency trade-off is traced *explicitly* as a Pareto frontier;
2) Adds a fleet-wide budget on the use of vulnerable residential streets, so a planner can cap network-level penetration of narrow neighbourhood roads even when individual routes are safe;
3) Calibrates per-edge crash probability through a fully documented public-source chain (MoRTH road-category fatality totals [1] + Mohan et al. 2017 [2] ordering constraints + an OpenStreetMap structural proxy), avoiding the need for the protected Integrated Road Accident Database (iRAD) microdata;
4) Replaces the linear lateness term commonly used in soft-time-window formulations with a convex exponential SLA penalty $\exp(\beta\tau) - 1$, which we approximate via piecewise-linear outer-envelope in the MILP and evaluate exactly in the heuristics;
5) Is implemented behind a unified solver/evaluation contract so that an exact MILP, a giant-tour-with-Bellman–Ford-split genetic algorithm, and an adaptive large neighbourhood search all evaluate solutions through identical $F_1$ and feasibility code, eliminating the "apples-to-pears" bias that plagues many VRP comparisons;
6) Uses real Blinkit dark-store coordinates extracted from a publicly committed KML snapshot, not synthetic uniform-disk depots, so the geometric difficulty of the instances reflects the actual topology operators face.

The remainder of the paper is organised as follows. Section II surveys related work in safety-aware routing and quick-commerce optimisation. Section III states the SA-VRPTW formulation in full. Section IV describes the BASM calibration chain, the BPR congestion model, and the dark-store data pipeline. Section V details the three solvers. Section VI reports the empirical study. Section VII discusses operator-facing implications, limitations, and threats to validity. Section VIII concludes.

## II. RELATED WORK

### A. VRPTW WITH SOFT TIME WINDOWS

The classical hard-time-window VRPTW [3] requires every customer to be served strictly inside a window $[e_i, l_i]$. The soft variant [4], [5] permits late arrival at a penalty, typically linear in lateness. Convex penalties have been proposed for SLA-driven settings [6] but their linearisation in MILP models is non-trivial and is usually omitted. We adopt $\exp(\beta\tau) - 1$ to reflect the contractual cost structure of q-commerce, where lateness becomes operationally expensive faster than linearly (rider re-routing, customer churn, SLA refunds).

### B. SAFETY-AWARE VEHICLE ROUTING

Hoseinzadeh et al. [7] model cumulative crash risk along a tour but use weighted-sum aggregation, which makes the safety/efficiency trade-off implicit and difficult to interpret. $\varepsilon$-constraint methods [8], [9] dominate weighted-sum approaches when the planner needs to read off a Pareto frontier rather than a single weighted optimum, and we adopt them here.

### C. QUICK-COMMERCE AND LAST-MILE ROUTING

Sinchaisri et al. [10] document the operational intensity of q-commerce dispatch in the United States. Field studies of Indian gig-delivery rider behaviour [11], [12] highlight a tension between SLA pressure and safe driving compliance. Our behavioural Monte Carlo (Section V-E) is calibrated against these findings, with a sensitivity sweep documenting how the conclusions move as the assumed compliance prior shifts.

### D. CRASH-RISK CALIBRATION IN ROUTING

Most safety-aware routing papers treat the per-edge crash probability as exogenous, with calibration relegated to a single citation or synthetic distribution. We take a different stance: every constant in our calibration file is traced to a public source, and we publish the derivation as a separate calibration note (Section IV).

## III. PROBLEM FORMULATION

### A. NOTATION

Let $G = (V, E)$ be the directed street graph for a city, extracted via OSMnx [13] with the drive network filter. Let $D \subset V$ be the set of depot nodes (k-means cluster centroids of the city's Blinkit dark stores, snapped to the nearest OSM nodes), and $C \subset V$ the set of customer nodes (sampled with residential-density weighting, constrained to a delivery radius around depots). Let $K = \bigcup_{d \in D} K_d$ be the rider fleet, partitioned by home depot.

For each street-graph edge $(i, j) \in E$:

- $\ell_{ij}$ is the length in metres,
- $v_{ij}^{\text{free}}$ is the free-flow speed (km/h, OSMnx imputed),
- $t_{ij}^{\text{free}} = 60\ell_{ij}/(1000 v_{ij}^{\text{free}})$ is the free-flow travel time in minutes,
- $\rho_{ij} = \alpha(V_{ij}/C_{ij})^{\beta}$ is the Bureau of Public Roads (BPR) [14] congestion factor with $\alpha = 0.15$, $\beta = 4$,
- $t_{ij} = t_{ij}^{\text{free}}(1 + \rho_{ij})$ is the congestion-inflated travel time used everywhere downstream,
- $r_{ij} \in [0, 0.99]$ is the per-traversal crash probability (BASM, Section IV-A),
- $h_{ij} \in \{0, 1\}$ indicates a residential street (OSM `highway` $\in$ {`residential`, `living_street`}).

For each pair $(i, j)$ with $i, j \in C \cup D$, the super-graph arc

$(i,j)$ is the shortest-time path on $G$, with cumulative quantities

$$T_{ij} = \sum_{e \in P_{ij}} t_e, \tag{1}$$

$$R_{ij} = \sum_{e \in P_{ij}} -\ln(1 - r_e), \tag{2}$$

$$H_{ij} = \sum_{e \in P_{ij}} h_e. \tag{3}$$

$R_{ij}$ is additive in log-survival because per-traversal crashes are treated as independent events along the path.

### B. CUSTOMER PARAMETERS

Each customer $i \in C$ has a demand $q_i \in \{1,2\}$, an order placement time $e_i \in [0,5]$ minutes, a promised arrival $\mathrm{ETA}_i = e_i + 10$ (the q-commerce ten-minute promise), and a service time $s_i \in \{2,4\}$ minutes (4 min for high-rise buildings, otherwise 2). Time windows are soft.

### C. DECISION VARIABLES

For rider $k \in K$ and arc $(i,j)$ in the super-graph:

- $x_{ij}^k \in \{0,1\}$: rider $k$ traverses arc $(i,j)$,
- $a_i^k \geq 0$: arrival time of rider $k$ at node $i$ (defined only when $i$ is visited),
- $w_i^k \geq 0$: idle wait at $i$ (early arrival),
- $\tau_i^k \geq 0$: lateness at $i$,
- $u_i^k \in \mathbb{Z}_{\geq 1}$: MTZ sequence variable for sub-tour elimination.

### D. OBJECTIVE

The primary objective $F_1$ minimises ETA-deviation plus convex SLA lateness penalty across the fleet:

$$F_1 = \sum_{k \in K} \sum_{i \in C} \left[ w_{\text{early}} \cdot w_i^k + \left( \exp(\beta_{\text{stw}} \tau_i^k) - 1 \right) \right]. \tag{4}$$

Travel time enters implicitly through the arrival-time linking constraints; congestion is already baked into $t_{ij}$; crash risk is handled as an $\varepsilon$-constraint, not as an objective term, so the operator can read its cost off the Pareto frontier.

### E. $\varepsilon$-CONSTRAINTS

Per-route crash-survival budget:

$$\sum_{(i,j)} R_{ij} x_{ij}^k \leq \bar{R}, \quad \forall k \in K. \tag{5}$$

Fleet-wide residential-edge budget:

$$\sum_{k \in K} \sum_{(i,j)} H_{ij} x_{ij}^k \leq \bar{H}. \tag{6}$$

Per-route residential cap:

$$\sum_{(i,j)} H_{ij} x_{ij}^k \leq \bar{H}_k, \quad \forall k \in K. \tag{7}$$

The pair $(\bar{R}, \bar{H})$ is swept on a $7 \times 6$ grid (Section VI) to trace the Pareto frontier; non-dominated points are retained.

### F. CORE VRPTW CONSTRAINTS

Each customer visited exactly once:

$$\sum_{k \in K} \sum_{j \in C \cup D, j \neq i} x_{ji}^k = 1, \quad \forall i \in C. \tag{8}$$

Flow conservation at every customer:

$$\sum_i x_{ij}^k = \sum_i x_{ji}^k, \quad \forall j \in C, \forall k. \tag{9}$$

Capacity:

$$\sum_{i \in C} q_i \sum_j x_{ij}^k \leq Q, \quad \forall k. \tag{10}$$

Multi-depot coupling: rider $k$ home-depoted at $d$ leaves $d$ at most once, returns iff it left, and cannot touch any other depot (omitted for brevity; full equations in the open-source repository).

Sub-tour elimination (MTZ):

$$u_i^k - u_j^k + |C| x_{ij}^k \leq |C| - 1, \quad \forall i,j \in C, \forall k. \tag{11}$$

Arrival-time linking:

$$a_j^k \geq a_i^k + s_i + T_{ij} - M(1 - x_{ij}^k), \quad \forall i,j,k. \tag{12}$$

Maximum route duration:

$$\sum_{(i,j)} T_{ij} x_{ij}^k + \sum_i s_i \sum_j x_{ji}^k \leq T_{\max}, \quad \forall k. \tag{13}$$

Soft time windows:

$$w_i^k \geq e_i - a_i^k, \quad w_i^k \geq 0, \tag{14}$$

$$\tau_i^k \geq a_i^k - \mathrm{ETA}_i, \quad \tau_i^k \geq 0. \tag{15}$$

### G. MILP LINEARISATION OF THE SLA PENALTY

The MILP cannot evaluate $\exp(\beta_{\text{stw}} \tau)$ directly. We approximate via a piecewise-linear outer envelope at breakpoints $\tau \in \{0, 2, 5, 10, 15, 20, 30\}$ minutes, encoded with SOS2 sets. The heuristics evaluate the exponential exactly. Both share the same feasibility checker and report $F_1$ through the same `savrptw.eval.objective` module.

Constants: $Q = 2$, $T_{\max} = 35$ min, $\bar{H}_k = 100$ residential edges per route, $\beta_{\text{stw}} = 0.12$, $w_{\text{early}} = 1.0$, $M = 10^4$. Per-depot rider counts are sized as $K_d = \lceil 1.2 N_d / Q \rceil$, providing a 20% over-provision buffer above the no-slack lower bound.

## IV. CALIBRATION AND DATA PIPELINE

### A. BENGALURU ACCIDENT SURROGATE MODEL (BASM)

The BASM converts publicly documented aggregate Indian road-safety rates into a transparent edge-level prior, which is then redistributed within each OSM road class using graph-local structure. The functional form is

$$r_{ij} = \mathrm{clip}\left[ \lambda_{\text{class}} \cdot \frac{\ell_{ij}}{1000} \cdot \sigma_{\text{class}} \cdot \pi_{ij}, 0, 0.99 \right], \tag{16}$$

where:

- $\lambda_{\text{class}}$ is a per-km, per-traversal base probability for the OSM road class, derived from MoRTH 2022 fatal+grievous events per category-km divided by the project's BPR-derived annual traversal exposure;

**TABLE 1. Blinkit dark-store coverage per city in the 2026-04-09 KML snapshot.**

| City | In-bbox count |
|---|---|
| Bengaluru | 113 |
| Delhi | 111 |
| Mumbai | 96 |
| Pune | 39 |
| Gurugram | 38 |
| Hyderabad | 0 (excluded) |

- $\sigma_{\text{class}}$ is a rank-preserving severity multiplier (arterial > collector > local), computed via the bounded transform $\sigma = \sqrt{\text{intensity}/g}$ where intensity is the fatality-share-to-length-share ratio per category and $g$ is the geometric mean of intensities;
- $\pi_{ij} \in [0.5, 2.0]$ is a bounded local proxy combining approximate edge betweenness, traffic-signal density, and crossing density at the edge endpoints.

The detailed derivation, including the page references for every MoRTH number used, is documented separately in the project's `BASM_CALIBRATION.md` note. We do not claim access to the protected iRAD microdata.

### B. BPR CONGESTION MODEL

$$\rho_{ij} = 0.15 \left( \frac{V_{ij}}{C_{ij}} \right)^4, \tag{17}$$

with capacity $C_{ij}$ from HCM 2010 [15] Exhibit 10-3 indexed by `(highway_class, lane_count)` and flow $V_{ij}$ derived from a class-level AADT prior multiplied by an hour-of-day factor (peak windows 08–11 and 18–21). All values are in `conf/congestion/bpr.yaml`.

### C. DARK STORES

We use the publicly available KML snapshot at https://darkstoremap.in/dark_store.kml, committed to the repository as `data/raw/darkstoremap_in_2026-04-09.kml`. The committed snapshot contains 461 Blinkit placemarks; per-city in-bounding-box counts are summarised in Table 1. Hyderabad is excluded from the primary single-brand experiments because the snapshot has zero in-bbox Blinkit placemarks there; refetching on 2026-04-25 produced a byte-identical file. For each retained city, we apply k-means clustering with $k \in \{3, 5, 8\}$ depots to obtain depot centroids.

### D. CROSS-VALIDATION AGAINST OPENSTREETMAP

We attempted to cross-validate the KML coordinates against OpenStreetMap by querying the Overpass API for features tagged `brand=Blinkit` or `name /Blinkit/i` within radii of 250 m and 1 km of each sampled coordinate. At a 1 km radius across 24 stores, only 2 matched (8%, 95% Wilson CI [0.02, 0.26]), driven by a single hit in Delhi and one in Gurugram. We interpret this as evidence about OSM's sparse coverage of Indian dark stores rather than about KML accuracy, and document this conclusion in the repository's `DARKSTORE_VALIDATION.md` note.

## V. SOLUTION METHODS

We implement three solvers behind a uniform contract: each consumes a typed `Instance` dataclass and returns a typed `Solution`. All three evaluate solutions through the same `savrptw.eval.objective.breakdown()` and `savrptw.eval.feasibility.validate()` functions, ensuring that reported $F_1$ values are strictly comparable.

### A. MILP

A PuLP/CBC model encodes the formulation in Section III directly. Two arrival-variable groups are kept separate to avoid a depot-arrival overload: $a_i^k$ for customer arrivals and $a_{\text{return}}^k$ for the home-depot return, which prevents the dispatch ($a_d^k = 0$) and return ($a_d^k > 0$) constraints from contradicting each other. The convex SLA penalty is linearised by SOS2-encoded outer-envelope segments at the breakpoints listed above. CBC is given a 120 s wall-clock budget per row in the default benchmark grid; longer budgets are needed only for $N \geq 50$ where the search is genuinely difficult.

### B. GENETIC ALGORITHM

The GA uses the giant-tour-with-Bellman–Ford-split decoder of Prins [16]: each chromosome is a permutation over $C$; the decoder finds the optimal partition into routes by solving a shortest-path problem on the auxiliary acyclic graph whose arcs correspond to feasible single-route cuts. Operators are OX1 crossover plus a randomised mutation choice over swap, insertion, and segment reversal (the last subsumes the 2-opt move). Tournament selection ($k = 5$), elitism (top 4 retained), default population 100, 200 generations.

### C. ADAPTIVE LARGE NEIGHBOURHOOD SEARCH

The ALNS implementation follows Ropke and Pisinger [17]. Destroy operators include random removal, worst removal (largest insertion-cost contribution), Shaw removal (relatedness over travel-time proximity), and a custom risk-cluster removal that targets customers in the highest-$R_{ij}$ neighbourhoods. Repair operators include cheapest-feasible insertion, regret-2, and regret-3. Operator weights are updated in segments of 100 iterations using the Ropke–Pisinger reaction factor; rewards are $\sigma = (33, 9, 1, 0)$ for new-best, improving, accepted-worse, and rejected moves. Acceptance is simulated annealing with a geometric cooling rate of 0.99975 starting from a temperature equal to 5% of the initial $F_1$. Algorithm 1 states the full loop.

**Algorithm 1. ALNS for SA-VRPTW**

**Input:** instance $\mathcal{I}$; iterations $T$; segment length $L$; reaction factor $r$; reward weights $\sigma$; cooling rate $\eta$; init. temperature ratio $\theta$
**Output:** feasible solution $s^\star$

**1.** $s_0 \leftarrow \mathrm{GreedyConstruct}(\mathcal{I})$
**2.** $s \leftarrow s_0$; $s^\star \leftarrow s_0$; $F^\star \leftarrow F_1(s_0)$
**3.** $T_{\mathrm{SA}} \leftarrow \theta \cdot F^\star$
**4.** $w_o^d, w_o^r \leftarrow 1$ for every operator $o$
**5. for** $t = 1$ **to** $T$ **do**
**6.** $q \leftarrow \lceil n \cdot \mathcal{U}(0.15, 0.45) \rceil$
**7.** $D \leftarrow$ destroy operator sampled $\propto w^d$
**8.** $R \leftarrow$ repair operator sampled $\propto w^r$
**9.** $s' \leftarrow R(D(s,q))$; $\Delta \leftarrow F_1(s') - F_1(s)$
**10.** **if** $F_1(s') < F^\star$ **then** $s^\star \leftarrow s'$; $F^\star \leftarrow F_1(s')$; $\rho \leftarrow \sigma_1$
**11.** **else if** $\Delta < 0$ **then** $\rho \leftarrow \sigma_2$
**12.** **else if** $\mathcal{U}(0,1) < \exp(-\Delta / T_{\mathrm{SA}})$ **then** $\rho \leftarrow \sigma_3$
**13.** **else** $\rho \leftarrow \sigma_4$
**14.** **if** $\rho > \sigma_4$ **then** $s \leftarrow s'$
**15.** $\pi_D^d \mathrel{+}= \rho$; $\pi_R^r \mathrel{+}= \rho$
**16.** **if** $t \bmod L = 0$ **then** $w_o^* \leftarrow (1-r)\, w_o^* + r\, \pi_o^* / n_o^*$; reset $\pi^*$ and counts
**17.** $T_{\mathrm{SA}} \leftarrow \eta \cdot T_{\mathrm{SA}}$
**18** **return** $s^\star$

### D. CRASH-SURVIVAL MONTE CARLO

For evaluation only (not optimisation), each route is sampled 10 000 times: each edge traversal is a Bernoulli trial with success probability $1 - r_e$, and the route survives iff every edge traversal succeeds. We report mean route survival, fleet survival, and Wilson 95% confidence intervals [18]. The analytic per-route survival $\exp(-R_{\mathrm{route}})$ is reported alongside as a sanity check.

### E. BEHAVIOURAL COMPLIANCE SENSITIVITY

Rider compliance is modelled as a $\mathrm{Beta}(\alpha, \beta)$ prior. The central scenario is $(\alpha, \beta) = (5, 2)$. We sweep $\alpha \in \{3, 5, 8\}$ and $\beta \in \{1, 2, 3\}$, yielding nine operating points. For each, we Monte-Carlo a 50-rider population over 5 000 trips and report the IQR band on per-rider survival, with edge-level traversal rates inflated by a factor $1 + k_c(1 - c)$ where $c$ is the sampled compliance level ($k_c = 0.4$ matched to Indian gig-rider field data [11]).

## VI. EXPERIMENTAL STUDY

### A. SETUP

We benchmark on five Indian cities (Bengaluru, Delhi, Gurugram, Mumbai, Pune) with depot counts $\{3, 5, 8\}$ scaled to $N$ as in Table 2. Customer counts are $N \in \{20, 50, 100, 200\}$. Each (city, $N$, solver) cell is replicated over 10 random seeds. The MILP is capped at $N \leq 50$ because CBC does not scale gracefully past that point inside the 120 s budget. Solvers are run on **[TBD: remote iDRAC compute node specification (CPU model, RAM, OS)]**.

**TABLE 2.** Per-instance depot counts as a function of $N$.

| $N$ | depots |
|---|---|
| 20 | 3 |
| 50 | 5 |
| 100 | 5 |
| 200 | 8 |

### B. REPRODUCIBILITY

The full pipeline (OSM graph → BPR congestion → calibrated risk → Blinkit dark-store clustering → super-graph construction → solver → evaluator) is composable through Hydra. A complete experiment row is reproducible from

```
PYTHONPATH=src python scripts/run_experiment.py
  city=bengaluru solver=ga instance.N=50 \
  instance.seed=42 instance.R_bar=0.40
```

and the full grid through `scripts/run_grid.py`, which checkpoints into a manifest so remote runs can resume from a crash without recomputing. Every parameter has a single source-of-truth file in `conf/`; the formulation note (`docs/FORMULATION.md`) and calibration note (`docs/BASM_CALIBRATION.md`) are versioned alongside the code.

### C. DEFAULT-$\varepsilon$ BENCHMARK

Table 3 reports mean $F_1$, mean wall-clock runtime, and feasibility rate per (city, $N$, solver) cell with $\bar{R} = +\infty$ and $\bar{H} = +\infty$ (no $\varepsilon$ constraints active). **[TBD: populate from data/results once the iDRAC remote run completes; the script `scripts/run_grid.py` is invoked with `-n-seeds 10 -solvers ga,alns,milp -milp-max-N 50`]**.

### D. PARETO FRONTIER: RISK VS. $F_1$

Figure 1 traces the $\bar{R}$-sweep at $N = 50$ for one seed per city. As $\bar{R}$ tightens from $\infty$ down to 0.05 the $F_1$ cost rises monotonically and the per-route survival probability $e^{-\bar{R}}$ correspondingly rises from **[TBD: ]** to **[TBD: ]**. The operator-facing reading: a one-percentage-point increase in mean per-route survival costs roughly **[TBD: ]** units of $F_1$ on average across the five cities.

### E. RESIDENTIAL-ROAD BUDGET SENSITIVITY

Tightening the per-route residential cap $\bar{H}_k$ from 100 to 50 edges costs **[TBD: ]**% in mean $F_1$ at $N = 50$ in Bengaluru and **[TBD: ]**% in Mumbai, with the gap explained by Mumbai's larger residential penetration coefficient **[TBD: ]**.

### F. SOLVER STATISTICAL COMPARISON

For each (city, $N$) cell, we compute the Wilcoxon signed-rank statistic [19] between paired GA and ALNS $F_1$ values across

**TABLE 3.** Default-$\varepsilon$ benchmark: mean $F_1$ (lower is better), mean wall-clock seconds, and feasibility rate over 10 seeds.

| City | $N = 20$ | | $N = 50$ | | $N = 100$ | |
|---|---|---|---|---|---|---|
| | MILP | GA | MILP | GA | ALNS | GA |
| Bengaluru | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |
| Delhi | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |
| Gurugram | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |
| Mumbai | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |
| Pune | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] | [TBD: ] |

**[Pareto plot — regenerate from data/results]**

**FIGURE 1.** Risk–$F_1$ Pareto frontier at $N = 50$. [TBD: regenerate once data/results contains the iDRAC sweep output.]

**[Behavioural sensitivity — regenerate from data/results]**

**FIGURE 2.** Behavioural-compliance sensitivity: per-rider survival vs. compliance prior parameters. [TBD: regenerate.]

the 10 seeds, applying Bonferroni correction across the resulting $5 \times 4 = 20$ comparisons. **[TBD: report the count of cells in which ALNS dominates GA at $\alpha = 0.05$ Bonferroni-corrected; expected pattern based on pilot runs is that ALNS dominates at $N \geq 100$.]**

### G. BEHAVIOURAL COMPLIANCE SWEEP

Figure 2 shows the mean per-rider survival as the compliance prior shifts from $(\alpha, \beta) = (8, 1)$ (high compliance) to $(3, 3)$ (low compliance). The dashed band is the IQR across the nine operating points. **[TBD: populate from data/results once `scripts/run_grid.py` has run with behavioural sensitivity enabled.]**

## VII. DISCUSSION

### A. OPERATOR-FACING IMPLICATIONS

The $\varepsilon$-constraint formulation lets a planner read the safety/efficiency trade-off as a tangible knob. Rather than tuning a weighted-sum coefficient until "it looks reasonable," the planner picks a target survival probability $e^{-\bar{R}}$ and the solver returns the cheapest fleet plan that meets it (or reports infeasibility). The fleet-wide residential-edge budget $\bar{H}$ gives a separate dial for political/community concerns about vulnerable-street penetration, which is operationally distinct from per-rider safety.

### B. LIMITATIONS

Three honest limitations remain. First, the BASM uses public aggregate data and an OSM structural proxy; it does not have access to the protected iRAD microdata, so the per-edge probability is a calibrated prior, not a measurement. Second, the dark-store snapshot cross-validation against OpenStreetMap returned $\sim$8% match at 1 km radius; this reflects sparse OSM brand tagging in India rather than a KML accuracy problem, but a manual sample audit on Google Maps would lift this from a coverage statistic to an accuracy figure. Third, MILP scaling to $N \geq 100$ is impractical with CBC; a commercial solver (Gurobi/CPLEX) would close some of that gap, but the heuristics already produce competitive bounds.

### C. THREATS TO VALIDITY

The customer placement model uses residential-density weighting on the OSM `landuse` layer rather than a population-density raster. Where census or WorldPop microdata is available, plugging it into `instance.generator` would tighten the demand model. The behavioural compliance prior is a single field-data citation, not a multi-study meta-analysis; the sensitivity sweep is the safeguard.

## VIII. CONCLUSION

We presented an SA-VRPTW formulation for q-commerce last-mile delivery in Indian cities that handles crash safety as an explicit $\varepsilon$-constraint, calibrates per-edge crash probability through a documented public-source chain, and is implemented under a unified solver/evaluation contract across MILP, GA, and ALNS. The empirical study on real Blinkit dark-store coordinates traces the Pareto frontier between $F_1$ delivery cost and per-route crash survival, and characterises the additional cost of network-wide residential-road budgets. The full pipeline is publicly released under the MIT licence.

Future work will lift three constraints. First, replacing the behavioural compliance prior with a multi-study meta-analysis. Second, incorporating real-time congestion feeds where available, replacing the BPR static model. Third, extending the formulation to a stochastic demand variant in

which order arrivals across the dispatch window are modelled by a non-stationary point process rather than a uniform $[0, 5]$ minute draw.

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